Unit 2 Summary: Analyzing One Variable

Name _____

Standard Deviation

Standard deviation	The SD is about how far the values usually are from the mean.
Small SD	The values are close to the mean. Player A scores close to 10 every game.
Large SD	The values are spread out. Player B might score 2 or 18.
Facts	The SD is never negative. It is 0 only when every value is the same. It has the same units as the data (points, minutes, inches).

Step	**Example: 7, 7, 10, 13, 13**
1. Find the mean, $\bar{x}$.	$\bar{x} = 50 \div 5 = 10$
2. Subtract the mean from each value.	$-3, -3, 0, 3, 3$
3. Square each result.	$9, 9, 0, 9, 9$
4. Add the squares.	$\Sigma(x - \bar{x})^2 = 9 + 9 + 0 + 9 + 9 = 36$
5. Divide by $n - 1$. This is the **variance**.	$36 \div 4 = 9$
6. Take the square root.	$s = \sqrt{9} = 3$
The formula: $s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}}$	Σ (capital sigma) means **add up**. $\Sigma(x - \bar{x})^2$ means: find $(x - \bar{x})^2$ for every value, then add them all. n is how many values there are. s is the standard deviation.

Choosing Measures

Shape of the data	**Center**	**Spread**
Symmetric, no outliers	mean	standard deviation
Skewed, or has outliers	median	IQR

Why: an outlier pulls the mean and the SD. It does not pull the median and the IQR much.

Comparing Two Distributions

	Ask	**Example: hours of sleep last night**
Shape	Symmetric or skewed?	9th grade is symmetric. 12th grade is skewed right.
Center	Which group is higher? (median or mean)	The median for 9th grade (8 hours) is greater than for 12th grade (6 hours). 9th graders usually sleep about 2 hours more.
Spread	Which group is more spread out? (IQR or SD)	Both groups have an IQR of 2 hours. The spreads are about the same.
Outliers	Any values far from the rest?	One 12th grader slept 12 hours. That is an outlier (the upper fence is 10).

Use comparing words: "The median for 9th grade is **2 hours greater than** for 12th grade." Use numbers and the context.

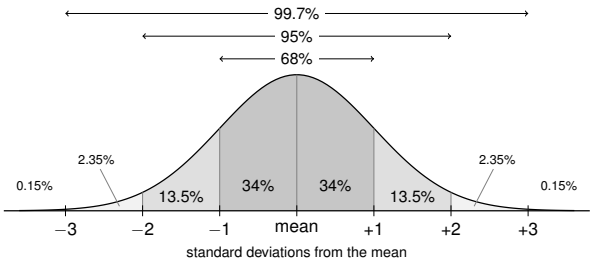
Percentiles

Percentile	**Example: 20 students took a test**
The percent of the data that is **below** the value. $\frac{\text{number of values below}}{\text{total number of values}} \times 100$	15 students scored below 85. $15 \div 20 = 0.75$. A score of 85 is at the **75th percentile**. It is higher than 75% of the class.
A percentile is **not** a percent score.	Scoring at the 75th percentile does **not** mean you got 75% of the questions right. It means you scored higher than 75% of the people.

Z-Scores

	What it tells	Example: mean 70, SD 10
Z-score $z = \dfrac{x-\mu}{\sigma}$	How many SDs a value is above or below the mean. μ (mu) is the mean. σ (small sigma) is the SD.	$z = \dfrac{85-70}{10} = 1.5$ 85 is **1.5 SDs above** the mean.
Sign of z	$z > 0$: above the mean. $z < 0$: below the mean. $z = 0$: equal to the mean.	55 has $z = -1.5$. It is 1.5 SDs **below** the mean.
Going backward	$x = \mu + z\sigma$	$z = -2$: $70 + (-2)(10) = 50$
Comparing	Z-scores let you compare values from **different** data sets.	Ana: 85, $z = 1.5$. Ben: 80 (mean 74, SD 3), $z = 2$. Ben did better compared to his class.

Normal Distributions and the Empirical Rule



Within	From	About
1 SD	$\mu - \sigma$ to $\mu + \sigma$	68%
2 SDs	$\mu - 2\sigma$ to $\mu + 2\sigma$	95%
3 SDs	$\mu - 3\sigma$ to $\mu + 3\sigma$	99.7%

A **normal distribution** is bell-shaped and symmetric. The mean and the median are in the center (they are equal). Two numbers describe it: the mean μ (center) and the SD σ (spread).

Example: heights, mean 64 inches, SD 3 inches	Answer
Percent from 61 to 67 inches	68% (within 1 SD)
Percent from 58 to 70 inches	95% (within 2 SDs)
Percent taller than 67 inches	13.5% + 2.35% + 0.15% = 16%
Percentile of 67 inches	100% − 16% = 84%, the 84th percentile

Changes: a larger SD makes the curve wider and flatter. A different mean moves the curve left or right.

Normal Calculations with a Z-Table

The z-table gives the area to the left of z. That is the proportion of the data below z.

Step	Example: heights
1. Draw the curve. Shade the part the question asks about.	Shorter than 68 inches: shade below 68.
2. Find each z-score: $z = \frac{x-\mu}{\sigma}$. Round to 2 decimal places.	$z = \frac{68-64}{3} = 1.33$
3. Look up each z-score in the table.	Row 1.3, column .03: .9082, so **90.82%**
4. **Below:** use the table. **Above:** 1− table. **Between:** subtract the two table values.	Taller than 68: 1 − .9082 = .0918, **9.18%** 62 to 68: $z = \frac{62-64}{3} = -0.67$, table .2514. .9082 − .2514 = .6568, **65.68%**

Normal Calculations in Reverse

Step	Example: heights, 90th percentile
1. Draw the curve. Shade the area to the **left**.	90% is to the left. The area is .9000.
2. Look **inside** the table for the area closest to it.	The closest area is .8997.
3. Read the z-score from the row and the column.	Row 1.2, column .08: $z = 1.28$
4. Find the value: $x = \mu + z\sigma$	$64 + 1.28 \times 3 = 67.84$, about **67.8 inches**
Bottom 10%: the area to the left is .10.	Closest area .1003, at $z = -1.28$. $64 + (-1.28)(3) = 60.16$, about **60.2 inches**
Top 10%: the area to the left is 1 − .10 = .90.	Subtract from 1 **before** you look in the table.